

Physiologically Based Pharmacokinetic Modeling to Assess Transporter-Mediated Drug–Drug Interactions in Support of Labeling Recommendations for Zavegepant Nasal Spray

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Zavegepant nasal spray, a calcitonin gene-related peptide receptor antagonist indicated for acute treatment of migraine with or without aura in adults, is a substrate for organic anion transporting polypeptide 1B3 (OATP1B3), Na⁺/taurocholate cotransporting polypeptide (NTCP), P-glycoprotein (P-gp), and cytochrome P450 3A4 (CYP3A4). In clinical drug–drug interaction (DDI) studies, oral zavegepant exposure increased in the presence of multiple-dose (MD) rifampin or itraconazole; however, no interaction was observed between intranasal (IN) zavegepant and itraconazole. A physiologically based pharmacokinetic (PBPK) model was developed and validated using *in vitro*, human mass balance, and observed clinical zavegepant pharmacokinetic data and was employed to assess clinical DDI mechanisms and further support zavegepant nasal spray labeling recommendations. PBPK modeling indicated that DDI observed between itraconazole and oral zavegepant is primarily driven by intestinal P-gp inhibition, and observed DDI with MD rifampin primarily results from hepatic uptake inhibition. Modeling predicted an increase in plasma exposure (AUC ratio [AUCR]=2.39) of IN zavegepant with single-dose (SD) rifampin; IN zavegepant plasma exposure was also predicted to increase with MD rifampin (AUCR=2.11). Predicted IN zavegepant exposure change with SD cyclosporine A was 1.58-fold (AUCR) when assuming 90:10 proportion for OATP1B3:NTCP contribution to hepatic uptake. CYP3A inducers, carbamazepine and efavirenz, were predicted to have minimal effect on IN zavegepant PK. Overall, PBPK modeling and observed PK data support that CYP3A/P-gp inhibitors and inducers do not produce clinically relevant modulation in IN zavegepant plasma exposure; however, OATP1B3/NTCP inhibitors can significantly increase IN zavegepant exposure, and coadministration should be avoided.

Study Highlights

WHAT IS THE CURRENT KNOWLEDGE ON THE TOPIC?

✓ Intranasal (IN) zavegepant is a prescription nasal spray used for the acute treatment of migraine headaches with or without aura in adults. Zavegepant is a substrate of OATP1B3, NTCP, and P-gp, with limited metabolism and oral availability.

WHAT QUESTION DID THE STUDY ADDRESS?

✓ Characterize the object drug–drug interactions (DDIs) of zavegepant and support labeling recommendations for zavegepant nasal spray via mechanistic physiologically based pharmacokinetic (PBPK) modeling and simulations.

WHAT THIS STUDY ADDS TO OUR KNOWLEDGE?

✓ The comprehensive mechanistic PBPK analysis including clinical PK data demonstrated CYP3A/P-gp inhibitors and

inducers do not produce clinically relevant modulation of IN zavegepant plasma exposure; however, OATP1B3/NTCP inhibitors can significantly increase IN zavegepant exposure, and coadministration should be avoided. This PBPK analysis also adds a valuable example of OATP1B3-predominant hepatic uptake to the limited body of published literature.

HOW MIGHT THIS CHANGE CLINICAL PHARMACOLOGY AND THERAPEUTICS?

✓ This study demonstrates that PBPK modeling and simulations can be valuable in investigating mechanisms of observed DDIs and support labeling recommendations in untested DDI scenarios.

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Zavegepant is a high-affinity small molecule antagonist of the calcitonin gene-related peptide (CGRP) receptor.¹ Zavegepant nasal spray (Zavzpret™; Pfizer Inc, New York, USA) is indicated in the United States for acute treatment of migraine with or without aura in adults.¹ The recommended dose is 10 mg as needed, with a maximum of one dose every 24 hours.¹ Intranasal (IN) administration allows for quick onset (peak plasma concentrations are observed ~30 minutes after a single dose) and may be beneficial to patients when oral treatment is not recommended because of nausea, vomiting, or tolerability concerns.¹

Zavegepant pharmacokinetic (PK) data in healthy participants are available from single ascending dose (SAD) and multiple ascending dose (MAD) studies following IN² or oral (data on file) administration, and a mass balance study with intravenous (IV) administration.³ Following IV administration of 5 mg [¹⁴C]-zavegepant, the majority of dose was recovered in feces as unchanged drug, with minor components of metabolism and unchanged renal excretion.³ Zavegepant IV clearance (CL_{iv}), steady-state volume of distribution (V_{ss}), and terminal half-life are 13.2 L/h, 41.6 L, and 6.8 hours, respectively.³ Following oral or IN administration, absolute bioavailability of zavegepant is 0.6%⁴ and 5%,^{1,4} respectively, and median time to reach maximum concentration (T_{max}) is 2 hours^{4,5} and 0.5 hours.^{1,4}

In vitro, zavegepant is a substrate for organic anion-transporting polypeptide 1B3 (OATP1B3; uptake transporter expressed in hepatocytes), Na⁺/taurocholate cotransporting polypeptide (NTCP; uptake transporter expressed in hepatocytes), and P-glycoprotein (P-gp; efflux transporter expressed in tissues including the intestine, kidney, and liver).^{1,6} Oxidative metabolism of zavegepant (a minor pathway of elimination) is primarily mediated by cytochrome P450 3A4 (CYP3A4) with minor contribution from CYP2D6.³

In a Phase 1 study in healthy participants, oral zavegepant exposure was increased in the presence of 600 mg once daily (QD) rifampin (strong CYP3A and P-gp inducer, OATP1B3 inhibitor, and weak NTCP inhibitor); area under the curve from time 0 extrapolated to infinity (AUC_{0-inf}) was increased by 2.3-fold, and maximal plasma concentration (C_{max}) was increased by 2.2-fold.⁵ Oral zavegepant exposure was also increased (AUC_{0-inf} by 1.6-fold and C_{max} by 1.8-fold) in the presence of 200 mg QD itraconazole (strong CYP3A and P-gp inhibitor); IN zavegepant exposure was not significantly changed in the presence of itraconazole.⁵ Overall, *in vitro* and *in vivo* findings suggest zavegepant drug–drug interactions (DDIs) are likely mediated primarily by OATP1B3, intestinal P-gp, and potentially NTCP.

Physiologically based pharmacokinetic (PBPK) models are increasingly used to predict DDIs in complex situations (e.g., simultaneous inhibition/induction of metabolic enzymes and transporters).^{7,8} Regulatory agencies have developed guidelines for PBPK modeling,^{9–11} and regulatory-approved instances of PBPK modeling to predict transporter-based DDIs and inform labeling recommendations have been published.⁷

The aim of this study was to apply PBPK modeling to characterize the object DDIs of zavegepant mechanisms and support labeling recommendations for zavegepant nasal spray. A PBPK model was developed based on *in vitro* and clinical PK data and extensively validated with clinical PK and DDI data. The model was subsequently adopted to simulate IN zavegepant exposure in clinically untested

DDI scenarios. Collectively, mechanistic modeling and simulations implicate the significance of OATP1B3/NTCP-mediated hepatic uptake as a major driver of object DDIs of IN zavegepant, whereas CYP3A and/or P-gp modulators have limited effects.

METHODS

Simcyp™ Population-Based Simulator (Version 20, Certara, Radnor, PA) was used for PBPK modeling and simulations. A systematic approach was adopted in the PBPK modeling with components of model development, model validation, and model application (Figure 1). As detailed below, a PBPK model was initially developed for zavegepant using physicochemical properties, *in vitro* studies, and clinical PK data following oral and IV administration. Observed clinical DDI data for oral zavegepant and itraconazole were used to estimate the contribution of P-gp and other efflux transporters in the intestine and liver, whereas the contribution of OATP1B3 to hepatic uptake was estimated from observed DDI data for oral zavegepant and rifampin. For IN dosing, a rapid first-order absorption and fraction of dose inhaled vs. swallowed were optimized to recover observed exposure in healthy participants. The PBPK model was validated against available clinical PK data following single- and multiple-dose oral and IN zavegepant, and additionally against clinical DDI data of IN zavegepant in the presence of itraconazole. The validated IN zavegepant PBPK model was then applied to simulate DDIs with rifampin, cyclosporine A (OATP1B3, NTCP, and P-gp inhibitor), carbamazepine (strong CYP3A and P-gp inducer), and efavirenz (moderate CYP3A inducer).

Fixed individual trial designs were used for model development and validation simulations with demographics (height, weight, age, and sex) and sample size matching the clinical studies; 10 virtual trials were generated to assess variability across groups. Simulated concentration–time profiles (visual check) and PK exposure parameters (AUC and C_{max}) of zavegepant were compared to observed data. During application of the model for DDI simulations, 10 virtual trials in 10 healthy participants (50% female) ages 20–50 years were generated. Simulations were performed in a virtual population library of healthy participants (Sim-healthy volunteers) using default SV-Itraconazole, SV-Rifampin, SV-Carbamazepine, and SV-Cyclosporine files in Simcyp without modification (unless described). Table S1 summarizes the trial designs used in model development, validation, and application.

Zavegepant PBPK model development

A full PBPK model was developed to describe mechanisms of zavegepant absorption, distribution, metabolism, and excretion (ADME; Figure S1) using available *in vitro* and clinical data (Table 1). The model included a mechanistic absorption model (Advanced Dissolution, Absorption and Metabolism [ADAM] model including intestinal efflux), whole-body distribution model, permeability limited liver (PerL) model for hepatic metabolism including transporter-mediated uptake and efflux, and renal clearance. Model development is described in more detail below.

Oral absorption modeling. The ADAM model was employed to capture passive absorption, intestinal efflux, and metabolism components.¹² Because the aqueous solubility of zavegepant is high (~20 mg/mL at pH 3–6.8), it is assumed to remain in solution throughout the gastrointestinal tract (data on file). Effective human jejunal permeability ($P_{eff,man}$) was estimated based on passive permeability in both Caco-2 (0.174×10^{-6} cm/s at 3–10 μM) and MDCKII-MDR1 (0.447 to 1.34×10^{-6} cm/s at 0.1–100 μM) cell monolayers (data on file). This low passive permeability *in vitro* corresponds to very low bioavailability observed for both oral (0.6%)⁴ and IN (5%)^{1,4} zavegepant formulations. The efflux ratio was 2.4 in MDCKII-MDR1 cells and was reduced to 1.08 in the presence of the P-gp inhibitor, valspodar, indicating that zavegepant is a substrate for P-gp (data on file). A mechanistic effective permeability (MechPeff) model was utilized to recover predicted jejunal $P_{eff,man}$ (0.0836×10^{-4} cm/s) and to calculate zavegepant regional permeability values in the remaining

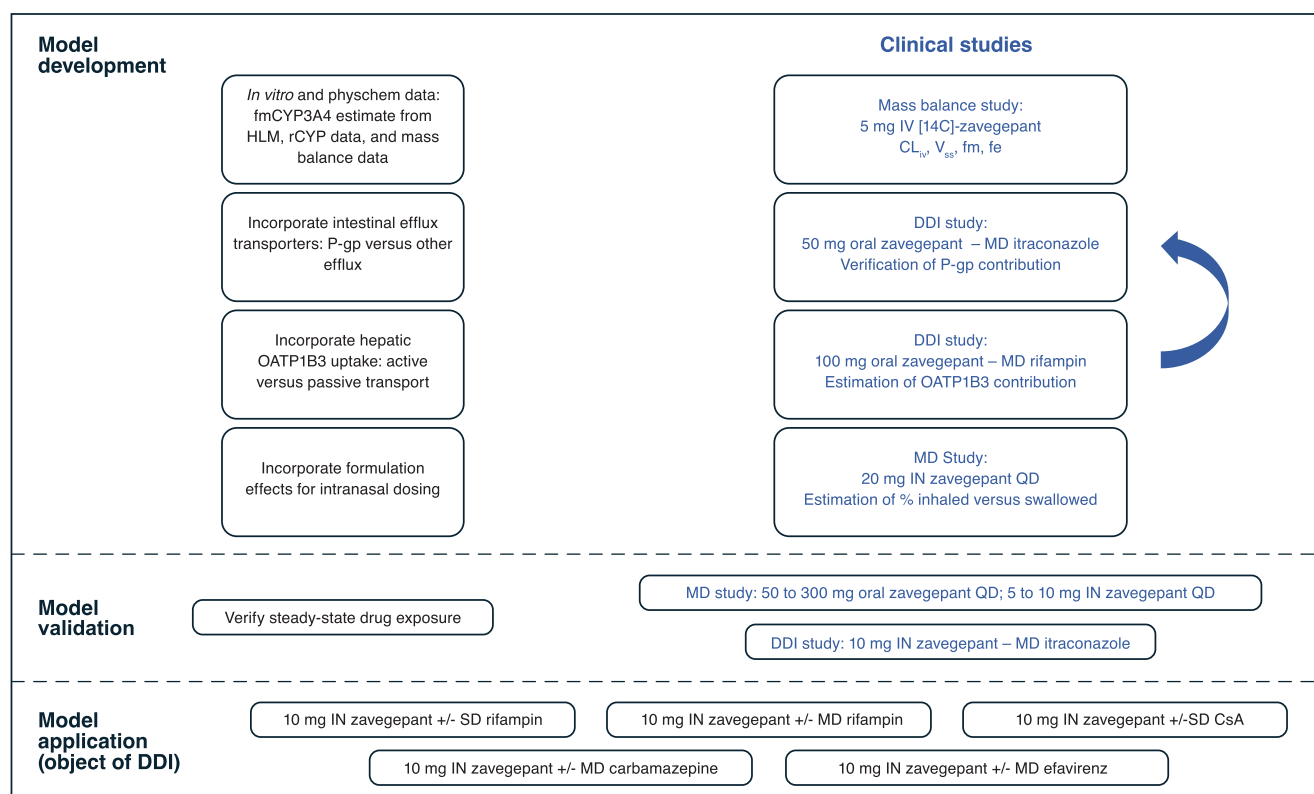


Figure 1 Key steps and clinical studies used in PBPK model building, validation, and application. All clinical studies and simulations were done in healthy participants. CL_{iv} , intravenous clearance; CsA, cyclosporine A; DDI, drug–drug interaction; f_e , fraction of drug excreted; f_m , fraction of drug metabolized; $f_{m,CYP3A4}$, fraction of drug metabolized by cytochrome P450 3A4; HLM, human liver microsomes; IN, intranasal; IV, intravenous; MD, multiple dose; OATP1B3, organic anion transporting polypeptide 1B3; PBPK, physiologically based pharmacokinetic; P-gp, P-glycoprotein; QD, once daily; rCYP, recombinant cytochrome P450; SD, single dose; V_{ss} , volume of distribution at steady state.

intestinal and colonic segments. Adjustment of intrinsic transcellular passive permeability values (reduction from 289 to 90×10^{-6} cm/s) was required within the MechPeff model to recover predicted jejunal $P_{eff,man}$. Colonic $P_{eff,man}$ was set to 0.001 to shift absorption to proximal parts of the gastrointestinal tract to match the observed T_{max} of 2 hours following oral dosing. An absorption rate scalar of three for Jejunum I was applied to recover observed C_{max} after oral zavegepant administration.¹³

Intestinal availability (F_g) was estimated to be high (0.99) given zavegepant's low metabolism. The fraction of zavegepant absorbed orally (F_a) was estimated from absolute oral bioavailability data after correcting for the fraction avoiding first-pass metabolism in the liver ($F_H = 0.83$) and gut F_g . Due to limitations in scaling *in vitro* efflux data, intestinal intrinsic P-gp clearance ($CL_{int,P-gp}$) (60 μ L/min) was optimized to recover an observed F_a of about 1% (Figure S2). Intestinal $CL_{int,P-gp}$ was further refined using sensitivity analysis by matching observed itraconazole DDI effects on 50 mg oral zavegepant (see below).

Distribution and elimination. For distribution, the Rodgers and Rowland model¹⁴ was initially applied. However, a global tissue-plasma partition coefficient (K_{pl}) scalar of 0.03 was required to better recover zavegepant IV data. Initial simulations failed to capture the terminal phase of the zavegepant PK profile. An additional organ compartment was used in the final model (with K_{p2} of 50) to capture the terminal phase of the zavegepant IV PK profile. Estimated V_{ss} of zavegepant (0.75 L/kg) with the final model was similar to observed V_{ss} .³

A clinically observed mean renal clearance (CL_R) of 1.22 L/h was input directly.² Observed mean zavegepant CL_{iv} was used in the retrograde model (extrapolation from *in vivo* data based on the well-stirred liver model) to estimate metabolic and biliary clearance.³ Hepatic metabolic CL_{int} was based on CL_{iv} and total fraction metabolized, and was further split between CYP3A4 (95%) and CYP2D6 (5%), based on *in vitro* phenotyping data (data on file). Total biliary intrinsic clearance was calculated, based on observed CL_{iv} and biliary fraction excreted, to be 15.7 μ L/min/million cells.³

Incorporation of intestinal and hepatic transporters. Zavegepant is a substrate for P-gp, OATP1B3, and NTCP transporters *in vitro*.^{1,6} Due to limitations of scaling *in vitro* transport data to *in vivo*, intestinal efflux and hepatic uptake components were incorporated in the PBPK model using sensitivity analysis to recover observed clinical DDIs.

Given the small contribution of CYP3A4-mediated metabolism to zavegepant clearance, DDI effects observed for multiple-dose (MD) itraconazole and oral zavegepant were attributed to inhibition of intestinal and hepatic P-gp.⁵ To refine the contribution of intestinal and hepatic P-gp, simulation of oral zavegepant (50 mg) plasma concentration–time profiles following QD administration of itraconazole in healthy participants was conducted. The Simcyp SV-Itraconazole_Fasted Soln and SV-OH-Itraconazole files were used, incorporating relevant P-gp competitive inhibition constant (K_i) values for itraconazole and OH-Itraconazole (0.00939 μ M and 0.096 μ M, respectively). The base zavegepant model assumed that efflux was solely mediated by intestinal P-gp. Sensitivity analysis showed that intestinal and hepatic efflux mediated by

Table 1 Final input parameters for the zavegepant PBPK model

	Value	Source
Physical chemistry and blood binding		
Molecular weight (g/mol)	638.82	Measured
Log P	2.6	Calculated from Log D, pKa 1, and pKa 2
Compound type	Diprotic base	Measured
pKa 1 and 2	4.8 and 8.8	Measured
Blood-to-plasma ratio	1.0	Measured
Fraction unbound	0.1045	Measured
Main binding protein	Albumin	Assumed
Absorption		
Model	ADAM	“Solution” option
Apparent permeability, $\times 10^{-6}$ cm/s	0.174	Measured
Calibrator apparent permeability constant (A->B) ($\times 10^{-6}$ cm/s) = atenolol	0.17	Measured
Effective human permeability ($P_{\text{eff,man}}$), Jejunum I, $\times 10^{-4}$ cm/s	0.0836	Predicted
Absorption rate scalar Jejunum I	3.0	Optimized based on matching observed C_{max}
Fraction unbound in the gut (F_{gut})	1.00	Default
Fraction absorbed (oral, F_a)	0.01	Simulated
Intestinal availability (F_g)	0.99	Simulated
Distribution		
Model	Full PBPK	-
Volume of distribution at steady state, Method 2, L/kg	0.75	Predicted
Tissue-plasma partition coefficient (K_{p1})	0.03	Optimized based on matching observed C_{max}
Additional organ tissue-plasma partition coefficient (K_{p2})	50	Optimized based on capturing terminal PK profile
Elimination		
Model	Recombinant	-
Intravenous clearance (CL_{iv}), L/h	15.78 ^a	Measured
Fraction metabolized by CYP3A4 ($f_{\text{m,3A4}}$)	95%	Measured
Fraction metabolized by CYP2D6 ($f_{\text{m,2D6}}$)	5%	Measured
CYP3A4 intrinsic metabolic clearance, $\mu\text{L}/\text{min}/\text{pmol}$ ($CL_{\text{int,3A4}}$)	0.0577	Calculated from systemic plasma clearance and fraction metabolized
CYP2D6 intrinsic metabolic clearance, $\mu\text{L}/\text{min}/\text{pmol}$ ($CL_{\text{int,2D6}}$)	0.0553	Calculated from systemic plasma clearance and fraction metabolized
Renal clearance, L/h (CL_{R})	1.22	Measured
Biliary clearance, L/h (CL_{bile})	12.72	Calculated from systemic plasma clearance and fraction of biliary clearance = 80.5%
Intestinal transport		
Intrinsic clearance for P-gp, $\mu\text{L}/\text{min}$ ($CL_{\text{int,P-gp}}$)	20	Optimized based on the observed oral bioavailability and itraconazole DDI
Intrinsic clearance for other efflux transporters, $\mu\text{L}/\text{min}$ ($CL_{\text{int,other}}$)	40	Optimized based on the observed oral bioavailability and itraconazole DDI
Relative activity factor/relative expression factor	1	Default value assumed
Hepatic transport		
Passive diffusional clearance (CL_{pp}), $\mu\text{L}/\text{min}/\text{millions}$	7	Optimized based on rifampin DDI
Intrinsic clearance for OATP1B3, $\mu\text{L}/\text{min}/\text{millions}$ ($CL_{\text{int,OATP1B3}}$)	28	Optimized based on rifampin DDI
Intrinsic clearance for P-gp, $\mu\text{L}/\text{min}/\text{millions}$ (hepatic $CL_{\text{int,P-gp}}$)	5.23	Optimized based on biliary clearance and itraconazole DDI.
Intrinsic clearance for other efflux transporters, $\mu\text{L}/\text{min}/\text{millions}$ (hepatic $CL_{\text{int,other}}$)	10.47	Optimized based on biliary clearance and itraconazole DDI.

ADAM, advanced dissolution, absorption, and metabolism; C_{max} , maximal drug concentration; CYP, cytochrome P450; DDI, drug–drug interaction; OATP1B3, organic anion transporting polypeptide 1B3; PBPK, physiologically based pharmacokinetic; P-gp, P-glycoprotein; PK, pharmacokinetic; pKa, acid dissociation constant.

^aValue based on preliminary data. Published value is 13.2 L/h. Difference is not expected to impact PBPK modeling results.

P-gp vs. other unknown efflux transporters (1:2) was needed to recover observed itraconazole DDI effects on oral zavegepant.

The MechPeff model was used to incorporate the contribution of transporters and passive diffusion to hepatic uptake clearance. Due to inconclusive *in vitro* active and passive transport data, passive diffusional clearance (CL_{PD}) was initially estimated based on the MechPeff-based hepatocyte permeability. Hepatic uptake clearance mediated by OATP1B3 ($CL_{int,OATP1B3}$) was further optimized based on sensitivity analyses using oral zavegepant (100 mg) plasma concentration–time profiles without and with MD rifampin in healthy participants.

The Simcyp SV-Rifampin-MD file (Table S2) was used, incorporating relevant DDI mechanisms: P-gp inhibition ($K_i = 4.3 \mu\text{M}$), P-gp induction (maximal fold induction [Ind_{max}] = 3.5), and OATP1B3 inhibition ($K_i = 0.088 \mu\text{M}$). Default CYP3A4 induction parameters ($Ind_{max} = 16$ and $Ind_{C50} = 0.32 \mu\text{M}$) were also applied. The Simcyp SV-Rifampin model was qualified using clinical DDI data with established OATP1B1/1B3 substrates (e.g., rosuvastatin) and CYP3A substrates (e.g., midazolam, nifedipine), demonstrating acceptable predictive performance. These evaluations support the suitability of the rifampin PBPK model for assessing OATP1B3-mediated effects on zavegepant exposure. The base zavegepant model assumed a 50:50 contribution for active vs. passive transport. Sensitivity analyses were conducted by examining splits of 90:10 and 80:20 for active vs. passive transport to match observed DDI results with MD rifampin and oral zavegepant.⁵ Because rifampin is a weak inhibitor of NTCP ($K_i = 118$ – $188 \mu\text{M}$),^{15,16} NTCP-mediated transport was not incorporated in the model. MD rifampin DDI simulations did not incorporate any induction effect on OATP1B3 given the lack of definitive understanding in the literature.¹⁷

Intranasal absorption. IN absorption was captured using a first-order inhalation model with inputs including absorption rate ($k_{a,IN}$), fraction of dose inhaled, and 100% bioavailability of the inhaled drug. The percentage of dose inhaled vs. swallowed in humans was estimated by sensitivity analysis with the validated model using observed IN and oral bioavailability data and steady-state zavegepant exposure following 20 mg QD IN administration in healthy participants.^{1,4–6}

Model validation

The PBPK model was validated by comparing simulated and observed PK profiles and parameters following single and repeat QD administration of oral (50, 100, 200, and 300 mg; data on file) or IN (5 and 10 mg) zavegepant. The IN model and the minimal impact of itraconazole on IN zavegepant were further validated by comparing simulated and observed PK profiles and parameters following a single administration of 10 mg IN zavegepant in the absence of itraconazole and with steady-state dosing of 200 mg oral itraconazole.⁵ Simcyp SV-Itraconazole_Fasted Soln and SV-OH-Itraconazole files were used as described above.

Model application

The validated zavegepant PBPK model was used to simulate the DDI of a single IN dose of 10 mg zavegepant in the presence or absence of single-dose (SD) rifampin 600 mg, and also after 10 days of dosing with 600 mg QD rifampin. The Simcyp SV-Rifampin-SD file (Table S3) was used. To further elucidate the primary mechanisms for predicted MD rifampin DDI effects, additional simulations were performed with only the P-gp/CYP3A4 induction effect incorporated. The Simcyp SV-Rifampin-MD file was used as described above.

Simulations of a single IN dose of 10 mg zavegepant in the absence of, and coadministered with, SD cyclosporine 600 mg were then performed assuming the hepatic uptake of zavegepant was split 50:50 between OATP1B3 and NTCP. Simcyp Cyclosporine_Neoral (Table S4) and its metabolite SV-Cyclosporine_M17 (AM1; Table S5) files were used. The following relevant transporter inhibition constants were included: NTCP inhibition

($K_i = 0.37 \mu\text{M}$ for Cyclosporine A) and OATP1B3 inhibition ($K_i = 0.032 \mu\text{M}$ and $0.059 \mu\text{M}$ for cyclosporine A and AM1, respectively). Additional sensitivity analysis was performed with hepatic uptake of zavegepant split 90:10 between OATP1B3 and NTCP. The cyclosporine PBPK model file from the Simcyp library (version 20) was qualified using clinical DDI data with OATP1B1/1B3 substrates (e.g., repaglinide). Given cyclosporine's similar inhibition potency against both OATP1B1 and OATP1B3¹⁸ and the lack of OATP1B3 specific substrates, this model is suitable for assessing OATP1B3-mediated effects on zavegepant exposure.

Simulations of a single IN dose of 10 mg zavegepant administered in the absence of carbamazepine and administered on the 10th day of 11 days of dosing with 300 mg carbamazepine twice daily (BID) were also conducted. The Simcyp SV-Carbamazepine file was used to assess the effect of CYP3A4 and P-gp induction. Because carbamazepine is an inducer of both CYP3A4 and P-gp, sensitivity simulations were conducted where CYP3A4 induction alone (carbamazepine $Ind_{slope} = 0.16$, carbamazepine 10,11-epoxide metabolite $Ind_{slope} = 0.14$) and combined CYP3A4 and P-gp ($Ind_{max} = 1.75$) induction were incorporated. Finally, simulations of a single IN dose of 10 mg zavegepant in the absence of efavirenz and administered on the 10th day of 11 days of dosing with 600 mg efavirenz QD for Days 1–11 were conducted using the Simcyp SV-Efavirenz file with CYP3A4 induction ($Ind_{max} = 9.9$ and $Ind_{C50} = 3.8 \mu\text{M}$) incorporated.

RESULTS

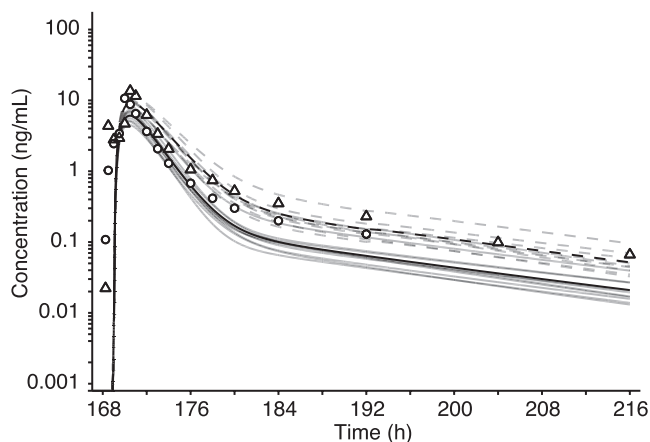
Zavegepant PBPK model development and refinement

Using the base model, the simulated plasma concentration–time profile of zavegepant following administration of a single 100 mg oral dose was comparable to observed clinical data⁵ (Figure S3), with predicted geometric mean values for AUC_{0-inf} and C_{max} within 0.80 to 1.25-fold of observed values (Table S6).

However, the base model, which assumes intestinal and biliary efflux primarily involves P-gp, overpredicted the effect of itraconazole on 50 mg oral zavegepant exposure. The model predicted AUC_{0-last} and C_{max} geometric mean ratio (GMR) values of 6.5 and 3.4, respectively, in comparison with observed values (1.74 and 1.77).⁵ Sensitivity analysis showed that a 1:2 ratio of P-gp vs. other efflux transporters in both the intestine and liver was needed to recover the observed DDI with itraconazole (Table S7). Using this 1:2 ratio, simulated profiles of 50 mg oral zavegepant in the presence and absence of 200 mg QD itraconazole were comparable to observed data (Figure 2a), with simulated AUC_{0-inf} and C_{max} GMRs for zavegepant within 0.80- to 1.25-fold of observed values⁵ (Table S6). The relative contribution of intestinal and hepatic P-gp vs. other efflux transporters was, therefore, set to 1:2 in the subsequent modeling.

The base model, which initially captured hepatic uptake with equal contribution (50:50) of active transport via OATP1B3 and passive uptake, underpredicted rifampin DDI with oral zavegepant. The base model predicted zavegepant AUC_{0-last} and C_{max} GMRs of about 1.25 and 1.35, compared to observed values of 2.32 and 2.17, respectively.⁵ The relative contribution between active uptake and passive transport in the liver was further refined to recover the rifampin DDI (Table S8). Overall hepatic uptake clearance was split 80:20 between active transport and passive diffusion in the final model, and estimations of CL_{PD} and $CL_{int,OATP1B3}$ were 7 and $28 \mu\text{L}/\text{min}/\text{million cells}$, respectively. This refined model well described the observed PK profiles of 100 mg oral zavegepant in the presence and absence of 600 mg QD rifampin (Figure 2b),

(a) 50 mg oral zavegepant ± 200 mg itraconazole QD



(b) 100 mg oral zavegepant ± 600 mg rifampin QD

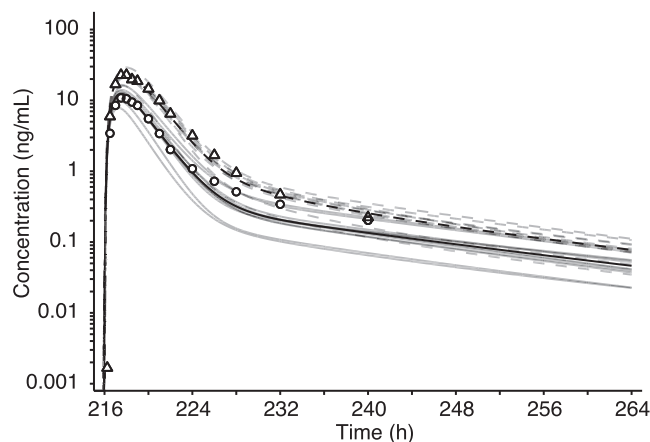


Figure 2 Simulated and observed zavegepant concentration–time profiles (model development) following a single oral dose of zavegepant administered alone and with MD itraconazole (a) or MD rifampin (b). Gray lines represent mean zavegepant concentrations of the simulated individual trials, and black lines represent mean concentration for the simulated population ($n=180$ for a; $n=150$ for b). Circles and triangles represent observed mean zavegepant concentrations ($n=18$ for a; $n=15$ for b). Solid lines/circles represent zavegepant concentration in the absence of the specified concomitant medication, and dashed lines/triangles represent concentrations in the presence of the concomitant medication. In a, 50 mg zavegepant was administered on the eighth day of 9 days of dosing with 200 mg itraconazole QD (simulations considered the inhibitory effect on CYP3A4, P-gp, and OATP1B3 by itraconazole and OH-itraconazole). In b, 100 mg zavegepant was administered on the 10th day of 10 days of dosing with 600 mg rifampin QD. CYP3A4, cytochrome P450 3A4; MD, multiple dose; OATP1B3, organic anion transporting polypeptide 1B3; P-gp, P-glycoprotein; QD, once daily.

with AUC_{0-inf} and C_{max} GMRs within 0.8- to 1.25-fold of the observed values⁵ (Table S6).

Model parameters for intranasal dosing. A 10% dose inhaled vs. 90% swallowed orally following IN dosing model best described steady-state drug exposure with 20 mg IN zavegepant for 14 days (Figure S4), with predicted geometric mean $AUC_{0-24,ss}$ and C_{max} values within 0.8- to 1.25-fold of observed values² (Table S9).

Zavegepant PBPK model validation

Prediction of oral and intranasal zavegepant pharmacokinetics. Using this middle-out model, simulated zavegepant concentration–time profiles following single and multiple (10 days) QD oral doses (50, 100, 200, and 300 mg) of zavegepant were comparable to observed data (Figure S5), with GMRs for AUC_{0-24} and C_{max} on Day 1 and Day 10 within 0.7- to 1.3-fold of observed values (data on file; Table S10). Similarly, model-predicted zavegepant concentration–time profiles following single and multiple (14 days) QD IN doses (5 and 10 mg) of zavegepant were comparable to observed data (Figure 3a,b), with GMRs for AUC_{0-24} and C_{max} on Day 1 and Day 14 within 0.7- to 1.4-fold of observed values² (Table S11).

Prediction of intranasal zavegepant DDI with itraconazole. Simulated concentration–time profiles following a single 10 mg IN dose of zavegepant administered in the absence of itraconazole and administered with dosing of 200 mg oral itraconazole QD to steady state were comparable to observed clinical data (Figure 3c). The predicted GMRs for AUC_{0-inf} and C_{max} in the presence of itraconazole were within 0.6- to 1.3-fold of observed values⁵ (Table S11).

Model application for predicting intranasal zavegepant DDIs with transporter and enzyme inhibitors/inducers

Intranasal zavegepant DDI with rifampin. Mean simulated plasma zavegepant concentrations, C_{max} , and AUC_{0-inf} are shown for a single IN dose of 10 mg zavegepant coadministered with SD rifampin 600 mg (Figure S6A) or with steady-state dosing of 600 mg rifampin QD (Figure S6B). Predicted zavegepant AUC GMRs were 2.39 and 2.11 with coadministration of SD or MD rifampin, respectively (Table 2). Sensitivity analysis estimated only a small (11%) reduction in zavegepant exposure due to P-gp/CYP3A4 induction (i.e., no OATP1B3 inhibition) of MD rifampin.

Intranasal zavegepant DDI with cyclosporine A. Mean predicted plasma zavegepant concentrations, AUC_{0-inf} and C_{max} following a single IN dose of 10 mg zavegepant in the absence and following 600 mg SD cyclosporine A are shown in Figure S6C. Predicted AUC GMR was 1.37 (Table 2). The exact contribution of OATP1B3 and NTCP to active hepatic uptake is unknown. Sensitivity analysis with hepatic uptake of zavegepant split 90:10 between OATP1B3 and NTCP predicted an AUC_{0-inf} GMR of 1.58 for IN zavegepant in the presence of cyclosporine A, which was not significantly different from the GMR of 1.37 when assuming 50:50 contribution by OATP1B3 and NTCP (Table 2).

Intranasal zavegepant DDI with carbamazepine and efavirenz. Mean simulated plasma zavegepant concentrations following a single IN dose of 10 mg zavegepant in the absence of carbamazepine and with steady-state dosing of 300 mg carbamazepine BID (assuming CYP3A4 induction alone and combined CYP3A4

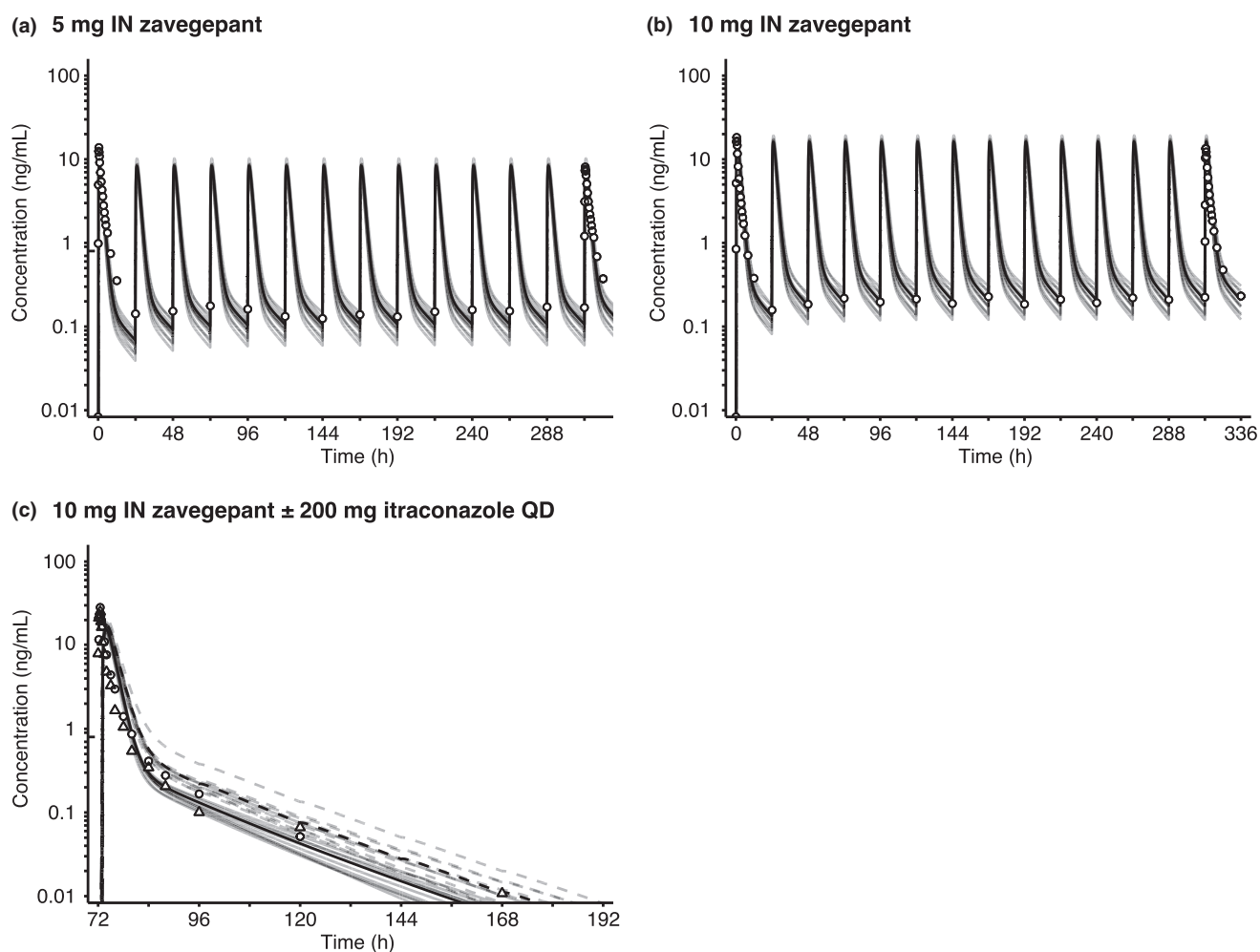


Figure 3 Simulated and observed zavegepant plasma concentration–time profiles (model validation) following 5 mg IN zavegepant (a), 10 mg IN zavegepant (b), and 10 mg IN zavegepant administered with MD itraconazole (c). Gray lines represent mean zavegepant concentrations of the simulated individual trials, and black lines represent mean concentration for the simulated population ($n=90$ for a and b; $n=180$ for c). Circles and triangles represent observed mean zavegepant concentrations ($n=9$ for a and b; $n=18$ for c). In c, solid lines/circles represent zavegepant concentration in the absence of the specified concomitant medication, and dashed lines/triangles represent concentrations in the presence of the concomitant medication. In c, 10 mg IN zavegepant was administered on the fourth day of 9 days of dosing with 200 mg itraconazole QD. IN, intranasal; MD, multiple dose; QD, once daily.

and P-gp induction) are illustrated in [Figure S6D,E](#). Predicted induction effects of carbamazepine on IN zavegepant were < 10% when considering CYP3A4 induction alone and combined CYP3A4 and P-gp induction ([Table 2](#)). Likewise, predicted CYP3A4 induction effects of efavirenz (600 mg QD) on IN zavegepant were < 10% ([Figure S6F](#), [Table 2](#)).

DISCUSSION

The aim of this study was to develop a PBPK model for zavegepant based on available *in vitro* and PK data to understand the mechanisms involved in observed clinical DDIs and to further support DDI labeling recommendations for zavegepant nasal spray, the approved formulation. Clinical studies in healthy participants showed increased oral zavegepant exposure with MD rifampin (strong CYP3A and P-gp inducer, OATP1B3 inhibitor, and weak NTCP inhibitor) and itraconazole (strong CYP3A and P-gp inhibitor).⁵ Exposure following administration of zavegepant

nasal spray was unchanged in the presence of MD itraconazole⁵ but was not evaluated in the presence of SD and MD rifampin, making it difficult to discern zavegepant nasal spray DDI potential with CYP3A4 and P-gp induction vs. OATP1B3 and NTCP inhibition. PBPK modeling suggests the observed DDI between oral zavegepant and MD itraconazole primarily is due to inhibition of intestinal P-gp, and the observed DDI with MD rifampin is mainly associated with inhibition of hepatic OATP1B3. Consequently, PBPK modeling predicted a lack of CYP3A4- and P-gp-mediated DDI risk following administration of zavegepant nasal spray ([Table 2](#), [Figure 4](#)). OATP1B3 and NTCP inhibitors, however, may elevate IN zavegepant exposure, and coadministration should be avoided. This mechanistic analysis was valuable in informing the product label on DDIs for zavegepant nasal spray.¹

A key challenge during model development was to incorporate intestinal efflux and hepatic uptake components. *In vitro* studies confirmed zavegepant is a substrate of OATP1B3, NTCP, and

Table 2 PBPK-predicted DDIs following a single dose of 10 mg intranasal zavegepant from simulations conducted during model application

Concomitant medication	GMR (90% CI)	
	AUC _{0-inf}	C _{max}
Rifampin		
SD 600mg rifampin (OATP1B3 inhibition)	2.39 (2.30, 2.48)	1.70 (1.65, 1.75)
MD 600mg rifampin (OATP1B3 inhibition, CYP3A4 and P-gp induction)	2.11 (2.03, 2.20)	1.67 (1.63, 1.72)
MD 600mg rifampin (CYP3A4 and P-gp induction only)	0.89 (0.89, 0.89)	0.99 (0.98, 0.99)
Cyclosporine A		
SD 600mg cyclosporine A (OATP1B3, NTCP, and P-gp inhibition; OATP1B3:NTCP = 50:50 ^a)	1.37 (1.34, 1.39)	1.13 (1.11, 1.15)
SD 600mg cyclosporine A (OATP1B3, NTCP, and P-gp inhibition; OATP1B3:NTCP = 90:10 ^a)	1.58 (1.55, 1.61)	1.23 (1.20, 1.25)
Carbamazepine		
BID 300mg carbamazepine (CYP3A4 and P-gp induction)	0.94 (0.93, 0.94)	0.98 (0.98, 0.99)
BID 300mg carbamazepine (CYP3A4 induction only)	0.98 (0.97, 0.98)	1.00 (1.00, 1.00)
Efavirenz		
QD 600mg efavirenz (CYP3A4 induction)	0.94 (0.94, 0.95)	0.99 (0.99, 0.99)

AUC_{0-inf}, area under the curve from time 0 to infinity; BID, twice day; CI, confidence interval; C_{max}, maximal drug concentration; CYP3A4, cytochrome P450 3A4; DDI, drug–drug interaction; GMR, geometric mean ratio; MD, multiple dose; NTCP, Na⁺/taurocholate cotransporting polypeptide; OATP1B3, organic anion transporting polypeptide 1B3; PBPK, physiologically based pharmacokinetic; P-gp, P-glycoprotein; QD, once daily; SD, single dose.

^aRatio represents relative contribution of OATP1B3 and NTCP to hepatic uptake of zavegepant in the simulation.

P-gp.^{1,6} However, due to limitations on scaling these *in vitro* transport data to *in vivo* parameters, PBPK model development utilized a middle-out approach while accounting for known disposition mechanisms and parametrization using a series of sensitivity analyses. Because the contribution of CYP3A4-mediated metabolism to zavegepant clearance is minimal,³ and no DDI was observed between IN zavegepant and MD itraconazole,⁵ observed DDI effects between itraconazole and oral zavegepant⁵ can be primarily attributed to inhibition of intestinal and hepatic P-gp. However, this assumption alone significantly overpredicted itraconazole-oral zavegepant DDI. Although it has been reported that hydroxypropyl-β-cyclodextrin (HP-β-CD) in itraconazole oral solution could result in the underestimation of DDI effect due to reduced drug

absorption via inclusion complex formation for some drugs,^{19,20} this effect was not observed when drugs were given 1 hour post itraconazole dose^{21,22} as HP-β-CD is largely cleared from the proximal intestine within 1–2 hours.¹⁹ Zavegepant was administered under this condition; therefore, significant effect of HP-β-CD on zavegepant absorption and underestimation of the true CYP3A4- and P-gp-mediated DDI in the clinical study of oral zavegepant with itraconazole oral solution is not expected. Hence, incorporation of other efflux transporters in addition to contribution of both intestinal and hepatic P-gp-mediated efflux was necessary. Potential for other efflux transporters cannot be confirmed with available *in vitro* experimental data. Additionally, confidence in quantitative prediction of intestinal efflux-mediated DDIs is relatively low, and the plausibility of itraconazole P-gp inhibitor parameters (*IC*₅₀) overestimating the magnitude of inhibition at the gut level cannot be ruled out.^{23,24}

Itraconazole showed an increase in plasma exposure of oral zavegepant but not following IN zavegepant, suggesting limited CYP3A-based hepatic metabolism and potentially low gut extraction (high *F*_g). The PBPK model well described these clinical observations. Although itraconazole can maximally inhibit gut P-gp activity, its effect on hepatobiliary P-gp efflux is expected to be low due to high protein binding and, thus, predicted to have limited effect on zavegepant biliary secretion. Additionally, hepatic uptake is often the rate-determining step in hepatic clearance of OATP1B substrate drugs, and metabolism and/or biliary secretion can have a limited impact on plasma exposure.^{25,26} Low-permeability OATP1B substrate drugs like rosuvastatin and fexofenadine, which are eliminated mostly via biliary secretion, are more sensitive to hepatic uptake modulation, with their oral exposure also influenced by gut efflux inhibition.^{27–29} Similarly, as zavegepant is an uptake substrate with low passive permeability, uptake modulators are expected to have a larger impact on its systemic clearance, which is evident with clinical DDI data and mechanistic modeling.

Rifampin-oral zavegepant DDI data were also utilized in refining the PBPK model, as the understanding on the contribution of active (OATP1B3) transport vs. passive diffusion to zavegepant hepatic uptake was limited based on preclinical data. Because rifampin is a strong inducer of CYP3A and P-gp, a decrease in oral zavegepant exposure is a possibility. However, oral exposure increased following MD rifampin treatment,⁵ suggesting an inhibitory effect on OATP1B3 that is greater than the induction effect on CYP3A4 and P-gp. Although rifampin transporter DDIs are commonly attributed to combined OATP1B1/1B3 inhibition, accumulating evidence indicates that OATP1B3 inhibition alone is also clinically relevant.²⁴ The endogenous OATP1B3 biomarker glycochenodeoxycholic acid 3-O-sulfate (GCDCA-S) exposure showed significant and consistent increases (13–16-fold AUCR) following single-dose rifampin administration across multiple studies compared to changes in coproporphyrin I/III as an OATP1B1 endogenous biomarker.^{30–32} Zavegepant shows a similar hepatobiliary disposition to GCDCA-S, characterized by hepatic uptake and predominant biliary elimination.³ In addition, zavegepant is a substrate of OATP1B3, not OATP1B1, therefore, rifampin's inhibitory effect on OATP1B1 does not affect

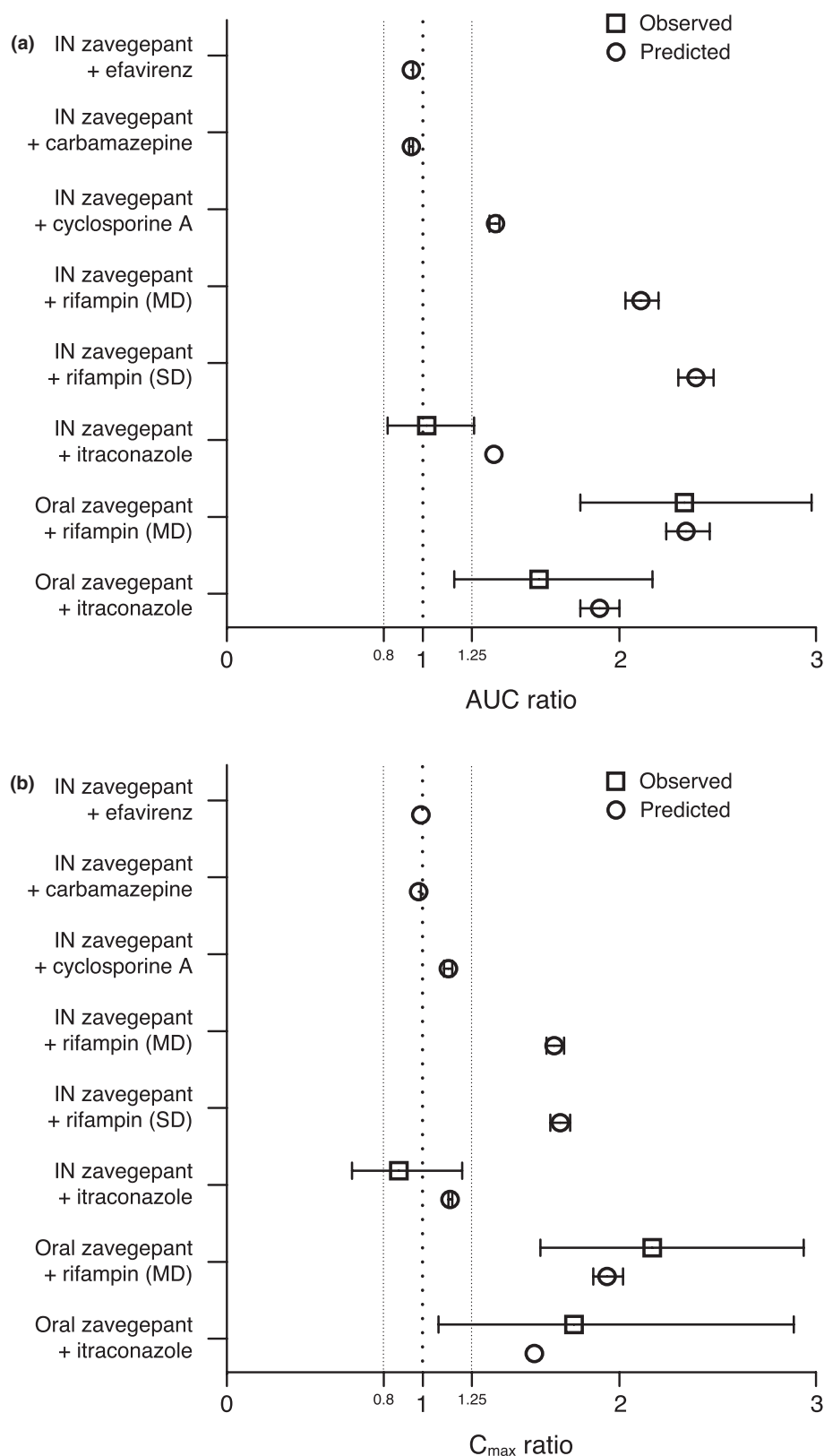


Figure 4 Observed and predicted DDI effects of CYP3A/P-gp and OATP1B/NTCP perpetrator drugs on IN and oral zavegepant $AUC_{0-\infty}$ (a) and C_{max} (b). Datapoints present GMR (90% CI). Dashed vertical lines represent bioequivalence limits of 0.8–1.25. AUC , area under the curve; CI, confidence interval; C_{max} , time to maximum plasma concentration; CYP3A, cytochrome P450 3A; DDI, drug–drug interaction; GMR, geometric mean ratio; IN, intranasal; MD, multiple dose; NTCP, Na+/taurocholate cotransporting polypeptide; OATP1B3, organic anion transporting polypeptide 1B3; P-gp, P-glycoprotein.

zavegepant PK prediction. Collectively, this evidence supports the significance of OATP1B3 inhibition in zavegepant-rifampin DDI. Use of the validated PBPK model to investigate potential DDIs following IN administration of 10 mg zavegepant predicted a 2.1-fold increase in exposure in the presence of MD rifampin. Therefore, similar AUC_{0-inf} changes with SD and MD rifampin (GMRs of 2.4 and 2.1, respectively, **Table 2**, **Figure 4**) imply a limited contribution of CYP3A/P-gp induction effects on zavegepant object DDIs. Because rifampin is not a potent NTCP inhibitor,^{15,16} hepatic NTCP-mediated transport was not initially incorporated in the PBPK model. Likewise, MD rifampin DDI simulations did not incorporate OATP1B3 induction effects, as clear clinical evidence on OATP1B3 induction by rifampin is lacking.³³ Additional MD rifampin simulations performed without incorporating OATP1B3 inhibition predicted a small reduction in zavegepant exposure, confirming DDI effects are primarily driven by OATP1B3 inhibition.

IN absorption was based on a first-order inhalation model, where fraction of dose inhaled vs. fraction swallowed orally was parameterized by fitting the validated PBPK model to observed IN PK. Oral zavegepant absorption is impaired (absolute bioavailability of 0.6%)⁴; therefore, an ~10-fold higher dose-normalized exposure for IN dosing compared with oral dosing justified the optimized contribution of fraction dose inhaled following IN administration. P-gp is expressed in human nasal mucosa,³⁴ though it is unclear if it exhibits functional activity in this context. Therefore, the IN absorption model did not account for efflux activity. Nonetheless, the lack of itraconazole effect on IN zavegepant implicates a limited role of P-gp efflux in the nasal mucosa, and/or exposures of inhibitors at the nasal absorption site are limited following oral dosing.

Using the PBPK model, clinical DDI scenarios with several OATP/NTCP inhibitors and CYP3A/P-gp inducers were simulated (**Table 2**, **Figure 4**). Predicted changes in zavegepant exposure in the presence of cyclosporine A (incorporating OATP1B3, NTCP, and P-gp inhibition) were moderate. Sensitivity analysis of OATP1B3 to NTCP contribution and inhibition potencies yielded <2-fold AUC_{0-inf} change in all scenarios. However, the robustness of the cyclosporine A model in predicting OATP1B3/NTCP substrates cannot be fully validated due to limited clinical DDI data. As expected from the small contribution of CYP3A4 metabolism to zavegepant clearance, predicted induction effects on zavegepant exposure in the presence of carbamazepine (strong CYP3A4 induction with and without consideration to P-gp induction) and efavirenz (moderate CYP3A4 induction) were small (<10% AUC_{0-inf} change). Overall, a significant DDI with OATP1B/NTCP inhibitors cannot be ruled out for IN zavegepant, whereas strong and moderate inducers of CYP3A and P-gp pose no DDI risk. The zavegepant FDA clinical pharmacology assessment⁶ recognized PBPK simulations as supportive evidence for interpreting DDI risk, particularly in distinguishing the relative contributions of CYP3A/P-gp vs. hepatic uptake transporter inhibition. This PBPK modeling helped contextualize the absence of clinically meaningful interactions with strong CYP3A/P-gp inhibitors/inducers for the zavegepant nasal spray formulation while supporting labeling recommendations to avoid concomitant

administration with OATP1B3/NTCP inhibitors. Integrating PBPK results with clinical DDI data contributed to labeling language regarding transporter-mediated DDI and the anticipated magnitude of exposure changes.

While PBPK modeling is routinely applied for DDI evaluation, this model represents a clinically validated example of OATP1B3-predominant hepatic uptake with minimal OATP1B1 contribution, a mechanism that is infrequently described in the literature. Additionally, the work demonstrates mechanistic differentiation between CYP3A/P-gp inhibitors/inducers and OATP1B3/NTCP inhibitors, leading to distinct regulatory outcomes (no restriction vs. contraindication) without the need for additional post-marketing clinical DDI studies. Zavegepant is a unique drug regarding disposition as reflected in the negligible oral absorption, nasal delivery, minimal metabolism, and hepatobiliary transport mechanisms, unlike commonly published examples involving CYP-based metabolism, and offers an interesting case example in the context of mechanistic modeling with insights of broader relevance beyond zavegepant.

A limitation of this analysis is the lack of selective and sensitive clinical OATP1B3 probe substrates to directly validate OATP1B3-mediated PBPK models. Consequently, the rifampin and cyclosporine PBPK models were evaluated primarily using substrates involving OATP1B1/1B3 for rifampin and OATP1B1 for cyclosporine. The rifampicin model was mainly validated by DDI with rosuvastatin, which is a substrate of both OATP1B1 and OATP1B3. Additionally, BCRP and other gut transporters may play a role in its interaction with rifampin. Based on clinical evidence and the employed (Simcyp v20) rosuvastatin model, OATP1B3 contribution to hepatic uptake is relatively lower than OATP1B1. Furthermore, the Simcyp (v20) default cyclosporine model is mainly validated against DDIs with repaglinide, which is an OATP1B1 substrate but not OATP1B3; therefore, OATP1B3-specific DDI translation is not directly validated. Mechanistically, since OATP1B1 and OATP1B3 are parallel pathways with inhibitors having similar *in vitro* inhibition potency against both transporters, the *in vivo* inhibition potency for both remains the same.³⁵ Therefore, validation of one pathway should be a good justification for the “intended use” of the PBPK model, particularly in the absence of selective substrates. Future studies with selective OATP1B3 probe substrates would further strengthen PBPK model validation for OATP1B3-mediated DDIs.

In conclusion, PBPK modeling and observed PK data support that CYP3A or P-gp inhibitors/inducers do not have clinically relevant effects on IN zavegepant exposure. OATP1B3 and NTCP inhibitors may significantly increase IN zavegepant exposure, and coadministration should be avoided. Collectively, this study offers a case example in implementing mechanistic modeling to retrospectively characterize the object DDI mechanisms involving transporters, leveraging existing clinical data, and prospectively predicting untested scenarios to enable labeling recommendations.

SUPPORTING INFORMATION

Supplementary information accompanies this paper on the *Clinical Pharmacology & Therapeutics* website (www.cpt-journal.com).

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CONFLICT OF INTEREST

Alice Ke (formerly) and Hannah M. Jones (currently) are employees of Certara and own stock/options in Certara. Ernesto Callegari, Manthena Varma, Mayur K Ladumor, Chieko Muto, Vaishali Sahasrabudhe, and Jing Liu are employees of Pfizer and own stock/options in Pfizer. Richard Bertz is an employee of Biohaven Pharmaceuticals and owns stock/options in Biohaven Ltd.

AUTHOR CONTRIBUTIONS

A.K., E.C., M.V.S.V., M.K.L., C.M., R.J.B., H.J., V.S., and J.L. wrote; R.J.B. and A.K. designed the research; A.K. performed the research; A.K. analyzed the data.

DATA AVAILABILITY STATEMENT

Upon request, and subject to review, Pfizer will provide the data that support the findings of this study. Pursuant to certain criteria, conditions, and exceptions, Pfizer may also provide access to the related individual de-identified participant data. For more information, go to <https://www.pfizer.com/science/clinical-trials/trial-data-and-results>.

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